

Problem

Let the line voltage $V_L = 208 \text{ V}$ and the wattmeter readings of the balanced system in Fig. 12.35 be $P_1 = -560 \text{ W}$ and $P_2 = 800 \text{ W}$.

Determine:

(a) the total average power

(b) the total reactive power

(c) the power factor

(d) the phase impedance

Is the impedance inductive or capacitive?

Step-by-step solution

Step 1 of 5

Given that,

$$V_L = 208 \text{ V}$$

$$P_1 = -560 \text{ W}$$

$$P_2 = 800 \text{ W}$$

(a) The total average power is ,

$$P_{\text{Total avg}} = P_1 + P_2$$

$$P_{\text{Total avg}} = -560 + 800$$

$$\therefore \boxed{P_{\text{Total avg}} = 240 \text{ W}}$$

Step 2 of 5

(b) The total reactive power is ,

$$Q_T = \sqrt{3}(P_2 - P_1)$$

$$Q_T = \sqrt{3}(800 - (-560))$$

$$\therefore \boxed{Q_T = 2355.6 \text{ VAR}}$$

Step 3 of 5

(c) The power factor is,

$$Pf = \cos \theta$$

Where,

$$\theta = \tan^{-1} \left(\frac{Q_T}{P_{Total\ avg}} \right)$$

$$Q = \tan^{-1} \left(\frac{2355.6}{240} \right)$$

$$\theta = 84.183^\circ$$

Hence,

$$Pf = \cos(84.183^\circ)$$

$$\therefore \boxed{Pf = 0.1014}$$

Step 4 of 5

(d) The phase impedance is,

$$Z_p = Z_p \angle \theta$$

We know that θ is the same as the Pf angle i.e.,

$$\theta = 84.183^\circ$$

$$Z_p = \frac{V_p}{I_p}$$

Where

$$P_p = V_p I_p \cos \theta$$

$$I_p = \frac{P_p}{V_p \cos \theta}$$

$$I_p = \frac{\left(\frac{1}{3} \times P_{Total\ avg} \right)}{\left(\frac{V_L}{\sqrt{3}} \right) (0.1014)}$$

$$I_p = \frac{\left(\frac{1}{3} \times 240 \right)}{\left(\frac{208}{\sqrt{3}} \right) (0.1014)}$$

$$\therefore I_p = 6.57 \text{ A}$$

Step 5 of 5

$$V_p = \frac{V_L}{\sqrt{3}}$$

$$V_p = \frac{208}{\sqrt{3}} \text{ V}$$

$$\therefore V_p = 120.1$$

Hence,

$$Z_p = \frac{120.1}{6.57}$$

$$Z_p = 18.28$$

$$\therefore \boxed{Z_p = 18.28 \angle 84.183^\circ}$$